

FINAL TEST, FALL 2024 Course: CS365 - Intro. to COA Duration: 60 minutes		Student name: Student Id: Order: Class Id:
Assessors' name and signature	Examiners' name and signature	Total score

Test Id: 1 This test consists of 10 questions. *No document is allowed.*

Questions with one correct answer

Question 1. What is the binary number corresponding to this hexadecimal number 0xC15?

- ☐ 111000010101 ☐ 110100010101 ☐ 101000010101 ☐ 110000010101

Question 2. What decimal number is represented by the following IEEE 754 single precision float?

1100 0001 0101 0110 0000 0000 0000 0000

- ☐ -12.375 ☐ -13.375 ☐ 13.375 ☐ 12.375

Question 3. Which of the following statements is correct?

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|---|---|
| ☐ Pipelining reduces the time it takes to complete an individual instruction | ☐ Pipelining reduces the number of simultaneously executing instructions |
| ☐ Pipelining improves instruction throughput | ☐ Pipelining improves individual instruction execution time |

Question 4. Which of the following terms is abbreviated as CPI?

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| ☐ Cycles per instruction | ☐ Clock per instruction |
| ☐ Clock cycles per instructions | ☐ Clock cycles per instruction |

Question 5. What is the Verilog code that creates a 32-bit value with the pattern 0101 ... 01?

- ☐ {2{16'b01}} ☐ {8'b1,4'b0} ☐ {16{2'b01}} ☐ 32'b01

Question 6. Which of the following statements is correct?

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|---|--|
| ☐ The output of a latch is not equal to the value of its stored state | ☐ DRAM is a memory where data is stored statically (as in flip-flops) |
| ☐ SRAMs are slower than DRAMs, but more dense and less expensive per bit | ☐ The pass transistor inside a DRAM's cell acts like a switch |

Constructed-response questions

Question 7. Add comments to the following MIPS code and describe in one sentence what it computes.

```
lwc1      $f4, 0($sp) .....
lwc1      $f6, 4($sp) .....
mul.s     $f2, $f4, $f6 .....
swc1      $f2, 8($sp) .....

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Question 8. Given the CPU performance equation:

$$\text{CPU time} = \text{Instruction count} \times \text{Average CPI} / (\text{Clock rate})$$

How does the instruction set architecture effect the factors in the CPU performance equation?

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Question 9. Give a brief introduction to your group project (i.e. the project that your group has taken in this course) and describe your contribution.

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Question 10. Draw the combinational logic circuit described by the following Verilog module.

```
module func {A, B, C, S, D}; .....
    input A, B, C; .....
    output S, D; .....
    assign S=A^B^C; .....
    assign D=(A&B) | (B&C) | (C&A); .....
endmodule .....

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Test Id: 2 This test consists of 10 questions. *No document is allowed.*

Questions with one correct answer

Question 1. What is the binary number corresponding to this hexadecimal number 0xBF4?

- ☐ 111011110100
 ☐ 101111110100
 ☐ 101011110100
 ☐ 110111110100

Question 2. What decimal number is represented by the following IEEE 754 single precision float?

1011 1111 0100 0000 0000 0000 0000

- ☐ -1.75
 ☐ -0.75
 ☐ 0.375
 ☐ -0.375

Question 3. Which of the following statements is correct?

- ☐ Pipelining reduces the time it takes to complete an individual instruction
 ☐ Pipelining reduces instruction throughput
☐ Pipelining increases the number of simultaneously executing instructions
 ☐ Pipelining improves individual instruction execution time

Question 4. Which of the following terms is abbreviated as GPU?

- ☐ Graphic processing unit
 ☐ Graphical processing unit
☐ General processing unit
 ☐ Graphics processing unit

Question 5. What is the Verilog code that creates a 32-bit value with the pattern 0101...01?

- ☐ {2{16'b01}}
 ☐ {16'b01,16'b01}
 ☐ {2{16'h5555}}
 ☐ 32'b01

Question 6. Which of the following statements is correct?

- ☐ The output of a flip-flop is not equal to the value of its stored state
 ☐ A CAM is a circuit that combines comparison and storage in a single device
☐ SRAM is a memory where data is stored dynamically
 ☐ DRAMs are faster than SRAMs, but less dense and more expensive per bit

Constructed-response questions

Question 7. Add comments to the following MIPS code and describe in one sentence what it computes.

```
lwc1    $f0, x($sp) .....
lwc1    $f1, y($sp) .....
div.s   $f2, $f0, $f1 .....
swc1    $f2, z($sp) .....

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Question 8. Given the CPU performance equation:

$$\text{CPU time} = \text{Instruction count} \times \text{Average CPI} / (\text{Clock rate})$$

Explain why optimizing the instruction set architecture can help to improve the CPU performance.

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Question 9. Give a short introduction to your group project (i.e. the project that your group has taken in this course) and your own contribution.

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Question 10. Draw the combinational logic circuit described by the following Verilog module.

```
module func {A, B, C, S, D}; .....
  input A, B, C; .....
  output S, D; .....
  assign S=A&B&C; .....
  assign D=(A^B) | (B^C) | (C^A); .....
endmodule .....

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```